CS19541 – COMPUTER NETWORKS LAB

PRACTICAL – 14

PACKET SNIFFING USING RAW SOCKETS

Aim

To develop a program with RAW sockets to capture network packets and show basic details about them.

Algorithm

1. Import Python's socket module.
2. Create a raw socket with the required address family and protocol.
3. Run the program with the necessary permissions.
4. Receive packets arriving at the system's network interface.
5. Print the packet's source information and total length.
6. Stop the capture process and close the socket.

Python Code – Windows

```
import socket

s = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
s.bind(("0.0.0.0", 0))
s.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)
s.ioctl(socket.SIO_RCVALL, socket.RCVALL_ON)
print("Packet sniffing started...")

try:
    while True:
        data, addr = s.recvfrom(65535)
        print("Packet received from:", addr[0])
        print("Packet length:", len(data))
        print("-" * 40)
except KeyboardInterrupt:
    print("\nSniffing stopped.")
finally:
    s.ioctl(socket.SIO_RCVALL, socket.RCVALL_OFF)
    s.close()
```

Input

Network packets that are visible to the selected system network interface.

Output

For every captured packet, the program displays the source IP address and packet length.

Note

This sample uses Windows-specific raw socket features and usually needs to be executed with Administrator privileges. Capture only network traffic you have permission to inspect.

Result

Thus, a basic packet-sniffing program using raw sockets was implemented to capture packets and display their information.